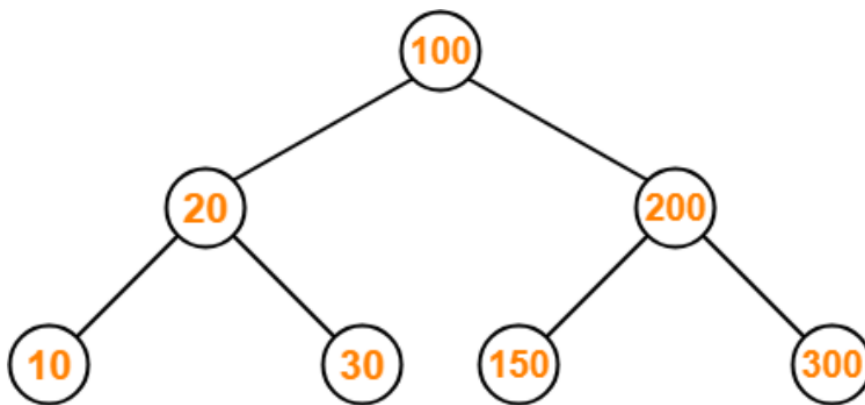


Write the traversal sequences for following trees

Question No :1



Tree: root 100, left child 20 (children 10, 30), right child 200 (children 150, 300).

**Preorder (Root, Left, Right):**

100, 20, 10, 30, 200, 150, 300

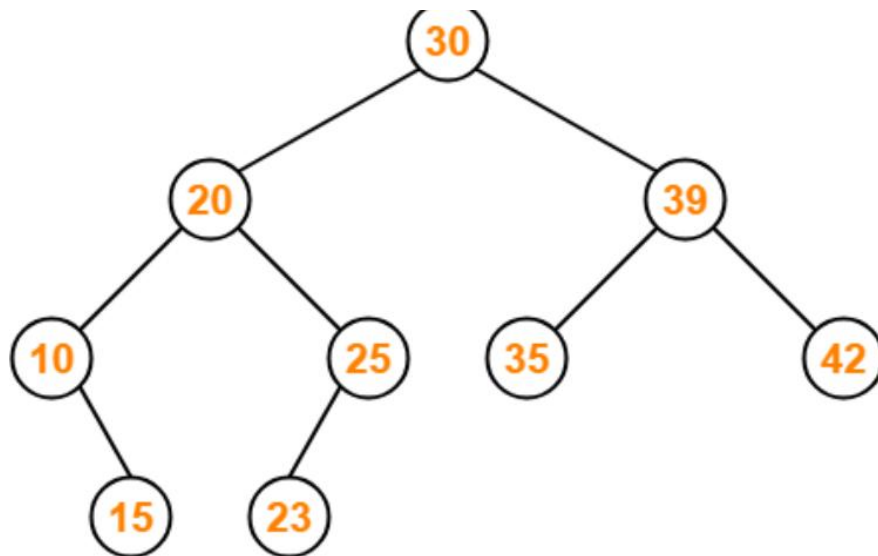
**Inorder (Left, Root, Right):**

10, 20, 30, 100, 150, 200, 300

**Postorder (Left, Right, Root):**

10, 30, 20, 150, 300, 200, 100

## Question No : 2



Tree: root 30, left child 20 (children 10, 25), right child 39 (children 35, 42); 10 has right child 15; 25 has right child 23.

**Preorder:**

30, 20, 10, 15, 25, 23, 39, 35, 42

**Inorder:**

10, 15, 20, 23, 25, 30, 35, 39, 42

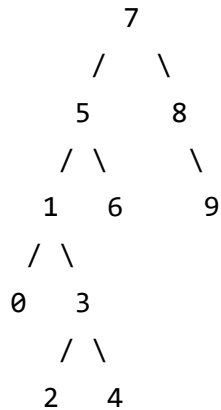
**Postorder:**

15, 10, 23, 25, 20, 35, 42, 39, 30

### Question No : 3

Assume the digits 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are added in that sequence into a binary search tree that is initially empty. The binary search tree follows the standard natural number ordering. What is the resulting tree's traversal sequence?

Inserting 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 in order into an initially empty BST gives the following structure:



**Preorder:**

7, 5, 1, 0, 3, 2, 4, 6, 8, 9

**Inorder (always sorted for a BST):**

0, 1, 2, 3, 4, 5, 6, 7, 8, 9

**Postorder:**

0, 2, 4, 3, 1, 6, 5, 9, 8, 7

**Question No : 5**



Tree: root 1, left child 2, right child 3; 2 has children 4 and 5; 3 has right child 6; 5 has left child 7.

**Preorder:**

1, 2, 4, 5, 7, 3, 6

**Inorder:**

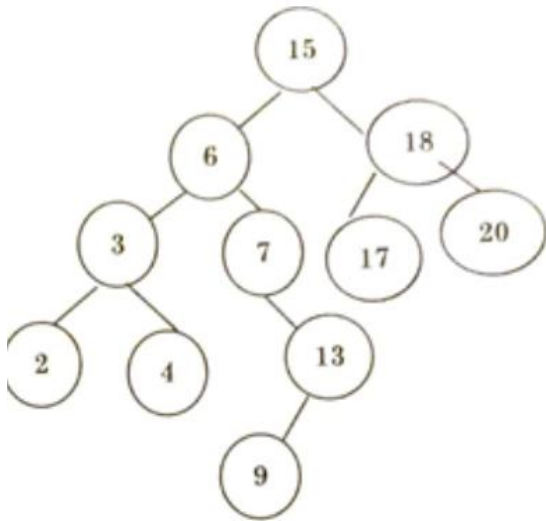
4, 2, 7, 5, 1, 3, 6

**Postorder:**

4, 7, 5, 2, 6, 3, 1

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### Question No 6



**Preorder:**

15, 6, 3, 2, 4, 7, 13, 9, 18, 17, 20

**Inorder:**

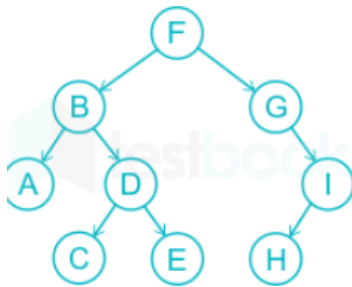
2, 3, 4, 6, 7, 9, 13, 15, 17, 18, 20

**Postorder:**

2, 4, 3, 9, 13, 7, 6, 17, 20, 18, 15

## Question No 7

What is the sequence of nodes when applying in-order traversal on the binary tree given below?



1. A, B, C, D, E, F, I, H, G

2. A, C, D, E, B, F, G, H, I

3. A, B, C, D, E, F, G, H, I

4. I, H, G, F, E, D, C, B, A

Inorder of B's subtree: A, B, C, D, E. Then root F. Then inorder of G's subtree: G, H, I.

**Answer: Option 3 — A, B, C, D, E, F, G, H, I**

### Question No 8

Consider the following Inorder and Preorder traversal of a tree

Inorder – D B E F A G H C

Preorder - A B D E F C G H

What is the Postorder Traversal of the same tree

1. D F E B H C G A

2. D F E B G H C A

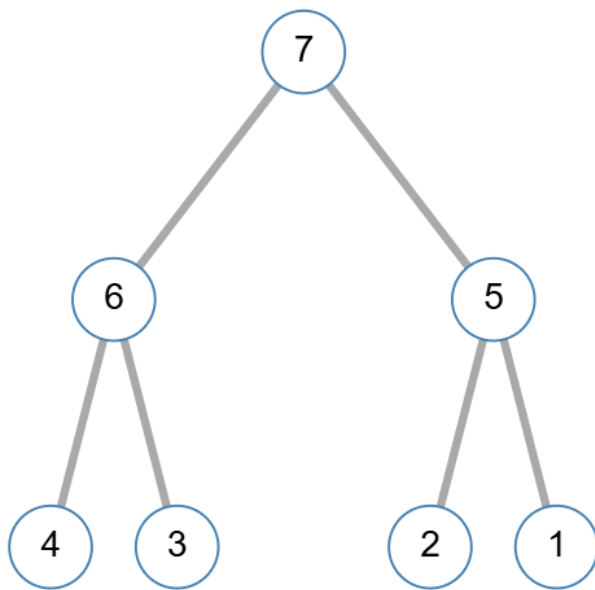
3. D F E B H G C A

4. D F E H B G C A

Reconstructing the tree from inorder + preorder gives: root A, left child B (B's left child D, B's right child E, E's right child F), right child C (C's left child G, G's right child H).

**Answer: Option 3 — D F E B H G C A**

### Question No 9 : Find Inorder, Preorder and Postorder



**Preorder:**

7, 6, 4, 3, 5, 2, 1

**Inorder:**

4, 6, 3, 7, 2, 5, 1

**Postorder:**

4, 3, 6, 2, 1, 5, 7

## Question No 10

This is a general (non-binary) tree: root A has children B, C, D. B has children E, F. F has children I, J, K. C has children G, H.

**Depth-First (Preorder) Traversal — visit root, then each subtree left to right:**

A, B, E, F, I, J, K, C, G, H, D

**Breadth-First (Level-order) Traversal — visit level by level:**

A, B, C, D, E, F, G, H, I, J, K

